MULTI-STEP PROMPT WORKFLOW

Mini Project: Task Scheduling System Using Priority Queue and Heap

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Scenario: Transforming an ADSD mini-project idea into a structured project proposal

1. Project Overview

Project Title: Task Scheduling System Using Priority Queue and Heap

The Task Scheduling System is an application that manages tasks based on their priority and execution requirements. The project uses advanced data structures such as Priority Queues and Heaps to efficiently organize, insert, remove, and execute tasks according to priority.

Suggested Technology: Java, Data Structures, Priority Queue, Heap

Users: Students, administrators, and system users

Core Features: Task entry, priority assignment, task scheduling, priority-based execution, task deletion, task search, and display of pending tasks.

2. Raw Project Notes

Many systems handle multiple tasks that need to be completed based on importance and urgency. Managing these tasks manually can cause delays and inefficient execution. The proposed Task Scheduling System should allow users to add tasks, assign priorities, automatically arrange tasks using a Priority Queue and Heap, execute the highest-priority task first, remove completed tasks, and view pending tasks.

STEP 1 — ANALYSIS & PROJECT PLANNING

Exact Prompt Used

Act as an expert Data Structures and Algorithms project analyst and technical documentation writer. I want to develop an ADSD mini project called 'Task Scheduling System Using Priority Queue and Heap'. Analyze the raw project notes below and produce:

- 1. The main problem statement.
- 2. Exactly 4 key project objectives.
- 3. The target users.
- 4. Exactly 6 core functional requirements.
- 5. The ADSD concepts and data structures used.
- 6. A structured outline for a 700-word mini-project proposal.

Do not introduce unrelated advanced technologies or unsupported features.

Output format: Problem Statement, Objectives, Target Users, Functional Requirements, ADSD Concepts Used, Proposal Outline.

Raw notes: Many systems handle multiple tasks that need to be completed based on importance and urgency. Managing these tasks manually can cause delays and inefficient execution. The proposed Task Scheduling System should allow users to add tasks, assign priorities, automatically arrange tasks using a Priority Queue and Heap, execute the highest-priority task first, remove completed tasks, and view pending tasks.

Intermediate AI Response — Live Execution Record

Paste the actual Step 1 response generated by the assigned AI interface here.

STEP 2 — PROPOSAL GENERATION

Exact Prompt Used

Excellent. Using only the project analysis, objectives, requirements, ADSD concepts, and proposal outline from your previous response, write a complete mini-project proposal for the Task Scheduling System Using Priority Queue and Heap.

Strict constraints:

- 1. Maximum 700 words.
- 2. Use a professional academic tone.
- 3. Include: Introduction, Problem Statement, Objectives, Proposed Solution, Key Features, ADSD Concepts Used, Expected Benefits, and Conclusion.
- 4. Explain how the Priority Queue and Heap work in simple terms.
- 5. Do not introduce technologies or features that were not included in Step 1.
- 6. Use short paragraphs of no more than 3 sentences each.
- 7. Include this simple text-based workflow: User → Task Input → Priority Queue/Heap → Task Selection → Execution → Updated Task List.
- 8. Make the proposal suitable for a college ADSD mini-project submission.

Output only the first draft of the proposal.

Intermediate AI Response — Live Execution Record

Paste the actual Step 2 response generated by the assigned AI interface here.

STEP 3 — CRITIQUE & REFINEMENT

Exact Prompt Used

Act as a strict senior Data Structures and Algorithms project reviewer. Critically evaluate the proposal you just wrote against the following checklist:

- 1. Is it 700 words or fewer?
- 2. Does it contain all required sections?
- 3. Are exactly 4 objectives represented?
- 4. Are the six core functional requirements accurately reflected?
- 5. Are Priority Queue and Heap concepts correctly explained?
- 6. Is the workflow clear and technically consistent?
- 7. Are the ADSD concepts relevant to the project?
- 8. Are unsupported or exaggerated claims avoided?
- 9. Are paragraphs limited to a maximum of 3 sentences?
- 10. Is the conclusion clear?

First, provide a 'Self-Critique' with a bullet-by-bullet evaluation of all 10 criteria. Identify any weaknesses and explain how they should be fixed.

Then provide 'Final Polished Mini-Project Proposal'. Correct every issue you identify while preserving the useful content from the previous draft.

The final proposal must remain within 700 words and satisfy all 10 criteria. Do not skip the self-critique.

Intermediate AI Response — Live Execution Record

Paste the actual Step 3 self-critique and final proposal generated by the assigned AI interface here.

4. Chaining Dependency

Step 1 → Step 2 → Step 3

Step 1 transforms the raw ADSD project notes into a structured analysis. Step 2 explicitly uses the output from Step 1 to generate a complete project proposal. Step 3 explicitly evaluates the Step 2 proposal against a defined checklist and transforms it into a corrected final version.

5. Critique Gate

Step 3 is the required critique gate. The AI must first review its own previous proposal, identify weaknesses against ten specific criteria, and then produce a refined final version that addresses the identified issues.

6. Live Execution Evidence

Run all three prompts sequentially in the same assigned AI interface. Add screenshots showing Prompt 1 and its response, Prompt 2 and its response, and Prompt 3 with the self-critique and final response. If the platform provides a shareable chat link, include it in the submission.

7. Assessment Checklist

- 3+ chained prompts — Three sequential prompts are provided.
- Explicit dependency — Steps 2 and 3 explicitly use the preceding outputs.
- Critique gate — Step 3 contains a 10-point review and correction stage.
- ADSD mini-project scenario — Task Scheduling System using Priority Queue and Heap is used.
- Live testing — Actual outputs and screenshots must be inserted after running the prompts.

8. Submission Note

Replace each 'Live Execution Record' placeholder with the exact AI-generated output obtained by running the three prompts in sequence. Add screenshots or the chat link as required by the assessment instructions.